1. What is a Robot?

Core Definition: An electromechanical system characterized by three essential traits:

- **Reprogrammable** Flexible software control
- **Multifunctional** Versatile multi-task capability
- **Sensible** Real-time environment awareness

Key Terminology:

- **Robot:** Physical mechanism performing human tasks autonomously or via teleoperation.
- **Robotics:** Science of mechanical design, kinematics, control & software.
- **Telerobotics:** Remote-controlled operation in extreme/hazardous environments.

2. Automation vs. Robots

Fixed Automation	Flexible Robots
Single dedicated task	**Diverse multiple tasks**
Maximum speed & efficiency	Reprogrammable motions
Rigid; high retooling cost	High mechanical flexibility
Ex: Bottling line, dishwasher	*Ex:* 6-DOF arms, AGVs, CNCs

3. Asimov's 3 Laws of Robotics

1. **Safety First:** May not injure a human or, through inaction, allow harm.
2. **Obedience:** Must obey human orders, **except** when conflicting with Law 1.
3. **Self-Defense:** Must protect own existence, unless conflicting with Laws 1 or 2.

4. Core Hardware Architecture

Sensors	**Data Input:** Internal state (encoders, resolvers) & external world (cameras, LIDAR, tactile/touch).
Controller	**The 'Brain':** Executes trajectories, computes kinematics & closes real-time feedback loops.
Actuators	**The 'Muscles':** Convert electrical/fluid energy into physical movement (servos, steppers, pneumatics).

5. Robot Types & Applications

Mechanical Architectures:

- **Manipulators:** Serial arms (Articulated, SCARA, Cartesian).
- **Mobile & Wheeled:** Differential drive, Omni, AGVs.
- **Legged:** Bipedal, quadrupeds, hexapods.
- **Autonomous Vehicles:** AUVs (Underwater), UAVs (Aerial drones).

Primary Drivers (3 D's / U's):

- **Dangerous** Nuclear cleanup, bomb disposal.
- **Dull / Repetitive** High-speed assembly, spot welding.
- **Dirty / Unwanted** Sewer inspection, toilet cleaning.

6. Robot Programming Methods

ON-LINE METHODS (At the Robot)

- **Teach Pendant:** Handheld button box. Jog arm to waypoints & save; controller calculates PTP paths.
- **Lead-Through:** Manually guide arm by hand. Records continuous stream (60–80 pts/sec; high memory).

OFF-LINE METHODS (Remote / Code)

- **Programming Languages:** Text code (AML, VAL, RobotStudio) without halting factory line.
- **Task-Level:** High-level goals ("pick item A") auto-compiled into trajectory commands.

7. Performance Metrics

- **Working Volume:** 3D envelope reachable by end-effector.
- **Speed & Accel:** Motion rate; trade-off with precision & payload.
- **Resolution:** Smallest commandable incremental step.
- **Accuracy:** Closeness between **commanded target** and **actual reached position** ($|x_{\text{target}} - x_{\text{actual}}|$).
- **Repeatability:** Ability to return to the **exact same point** over repeated cycles (σ).

HIGH-YIELD EXAM DISTINCTION

Accuracy vs. Repeatability: A robot can have **High Repeatability** (hits the identical spot 100 times in a row) but **Low Accuracy** (that spot is systematically 2 cm off target)!

1. Sensing & Transduction

- **Sensing:** Collecting information about the world.
- **Sensor:** Device mapping physical/chemical phenomena to quantitative signals.
- **Transduction Principle:** Converting one energy form into another.

Sensor Type	Transduction Mechanism
Thermistor	Temp → Resistance (R)
Photodiode	Light → Current (I)
Pyroelectric	Thermal rad. → Voltage (V)
Humidity	Moisture → Capacitance (C)
LVDT	Position → Inductance (L)
Microphone	Sound press. → Elect. signal

2. Sensor Classifications

State Focus:

- **Proprioceptive** (Internal): Measures internal parameters (joint angle, wheel position, battery, tachometers/encoders/accelerometers).
- **Exteroceptive** (External): Environment & objects (proximity, vision, range).

Energy Interaction:

- **Active:** Emits energy & reads reflection (Sonar, Radar, Modulated IR, LiDAR).
- **Passive:** Receives ambient energy only (Cameras, Pyroelectric, LDR).

Contact & Modality:

- **Contact / Non-contact:** Touch (bumpers) vs. remote (optical, ultrasonic).
- **Visual / Non-visual:** Vision cameras vs. non-optical sensors.

3. Contact & Resistive Sensors

A. FEELERS (TACTILE/CONTACT)

- **Whiskers:** Piano wire in metal hoop; deflection closes circuit; binary ($0/1$) output.
- **Bumpers & Guards:** Frame on microswitches; collision triggers switch; binary ($0/1$) output.

B. RESISTIVE (VARIABLE R)

- **Bend Sensors:** Flexible strips where R increases as bent ($10\ \mathrm{k\Omega} - 35\ \mathrm{k\Omega}$). *Uses:* Joint angles, wall-following, loads.
- **Potentiometers:** Linear/rotational variable resistors tracking sliding parts or rotating joint shafts.
- **Photocells (CdS / LDR):** Resistance shifts with light; highly non-linear, ideal for light tracking.

4. Capacitive vs. Inductive Sensors

Capacitive (All Materials)	Inductive (Metals Only)
• Measures change in **dielectric** • Detects **metals & non-metals** (liquids, plastics, powders) • Senses through glass/plastic • Sensitive to dust & humidity	• Emits AC magnetic field; senses **eddy currents** • Detects **metals ONLY** • Max on **ferrous** (iron/steel) • Immune to oil, dust & water
APPLICATIONS	**APPLICATIONS**
• **Level Detection:** Liquid/grains • **Proximity:** Assembly lines • **Touch Sensing:** Touchscreens	• **Position:** Metal part automation • **Speed:** Rotating gears/shafts • **Metal Detection:** Safety

5. Infrared (IR) Sensor Types

Type	Principle & Trade-offs	Applications
Intensity-Based *Reflective*	• Measures reflected light • **Pro** Cheap, simple • **Con** Ambient & color sensitive	• Line-following • Wall-tracking • Break-beams
Modulated IR *Proximity*	• Flashes at $32 - 45\ \mathrm{kHz}$ • Receiver demodulates signal • **Pro** Immune to ambient light	• Proximity sensing • Remote controls • Obstacle avoidance
IR Triangulation *PSD Ranger*	• Emitter + lens + **PSD** • Angle of return gives range • **Pro** Color & light immune	• Sharp GP2D02 ($10 - 80\ \mathrm{cm}$) • Distance map

6. Ultrasonic Sensors & Noise

- **Principle:** Ultrasound burst ($\approx 50\ \mathrm{kHz}$). Time-of-Flight (t):

$$D = v \cdot t \quad \Longrightarrow \quad d = \frac{v \cdot t}{2} \quad (v \approx 340\ \mathrm{m/s\ in\ air})$$

- **Limitations:**
  - **Bearing Uncertainty:** Wide beam spread (opening arc $\approx 30^{\circ}$).
  - **Speed Latency:** Sound is slow (e.g. $30\ \mathrm{m}$ round-trip takes $\approx 200\ \mathrm{ms}$).
- **Noise Issues:** Acoustic noise & sensor crosstalk.

7. Laser Range Finders (LiDAR)

- **Performance:** Long range ($2\ \mathrm{m} - 500\ \mathrm{m}$), high resolution ($10\ \mathrm{mm}$), fast scans ($13 - 40\ \mathrm{ms}$).
- **Robustness:** Narrow beam; immune to ambient light & specular reflections.

HIGH-YIELD EXAM COMPARISON

Inductive vs. Capacitive: Inductive senses **metals only**; Capacitive senses **all materials** (dielectric shift).
Ultrasonic vs. LiDAR: Sonar has wide **bearing uncertainty** ($\approx 30^{\circ}$) & latency; LiDAR provides millimeter angular precision.

1. Agency & Marr's 3-Level Theory

- **Agency:** Capacity of an entity (human or robot) to act intentionally, make choices, & exert control over actions and outcomes.
- **David Marr's Framework:** Analyzes information systems across 3 empirical levels:

Level	Question / Focus	Search & Rescue Robot Example
L1: Comp.	What goal / problem is solved?	Locate trapped survivors in low-visibility
L2: Algo.	What processes / IO represent it?	Steer toward heat & CO_2 gradients
L3: Impl.	How physically realized in hardware?	Angle vectors to heat centroids; wheel motors

KEY EXAM CONCEPT L1 & L2 are abstract & identical across biology/robots. Distinction appears **only at Level 3**.

2. Bio-Inspired Robotic Mappings

Biological Model	Robotic Mapping	Mechanism
Bat Echolocation	Obstacle Avoidance	Echo delay detection (ultrasonic)
Ant Foraging	Swarm Pathfinding	Ant Colony Optim. (ACO), digital pheromones
Bird Flocking	Coordinated Drone Swarm	Reynolds Boids: alignment, cohesion, separation
Frog Tongue	Fast Grasping Arm	High-speed camera + snatch reflex
Cuttlefish	Adaptive Camouflage	Pattern mapping onto E-skin

3. Classifications of Behavior

BEHAVIOR DEFINITION: Direct mathematical mapping of **sensory inputs** to **motor actions** to achieve a task.

1. Reflexive (Stimulus-Response) — Hardwired / Fastest

- **Reflex** — Lasts **only during stimulus** (knee-jerk, pupil reflex).
- **Taxis** — **Directional steering** relative to stimulus:
  • *Tropotaxis:* Turtles to moonlight | - *Phototaxis:* Worms away from light | + *Rheotaxis:* Salmon upstream
- **FAP** — Response **persists longer** than stimulus (spider web, nut caching, duckling imprinting).

2. Reactive Behaviors — Muscle Memory

Learned skills consolidated to execute automatically without conscious thought (e.g. riding a bicycle).

3. Conscious Behaviors — Deliberative

Deliberate, goal-directed planning coordinating multiple sub-behaviors (e.g. puzzle solving, kit assembly).

4. Behavioral Schema Architecture

Behavioral Schema: feed

eyes ⇒ **Perceptual Schema** "small, moving" (feature filter) — location, intensity ⇒ **Motor Schema** "snap" (action calc) — vector ⇒

- **Frog Case Study:** Raw visual stimulus (eyes) is filtered into a clean percept $(\mathrm{small,\ moving})$, passing coordinate (x, y) and intensity to the motor schema to generate a tongue **snap vector**.

5. Schema Theory & OOP Robotics

- **Schema:** Foundational OOP template used for the **Reactive Layer** of autonomous robots.
- **Components of a Behavior Schema:**
  - **Perceptual Schema** Sensory feature extraction, threshold filtering & stimulus delays.
  - **Motor Schema** Computational procedure to produce physical action/movement vector.
- **Schema Instantiation (SI):** Creating a concrete runtime instance from a generic parameterized class template (e.g. applying specific handlebar height & seat position to a bike-riding schema).

6. Vector Action & S-R Notation

Mathematical Formulations:

$$\{B : S \to R\} \quad \text{or} \quad B[S] = R$$

- S: Perceptual function converting raw sensor feeds to **percept**.
- R: Motor function converting percept into physical **action**.
- B: Overarching behavioral system.

Vector Field Combination:

- Directional stimuli (e.g. left & right eyes) combine via **Vector Summation** (Σ):

$$V_{\text{resultant}} = v_{\text{left}} + v_{\text{right}}$$

EXAM HIGHLIGHTS: LECTURE 03

Reflex vs. FAP Duration: Reflex stops as soon as stimulus stops; FAP continues for longer duration.
Taxis Examples: Positive Rheotaxis (salmon upstream), Tropotaxis (sea turtles to moonlight), Negative Phototaxis (earthworms).
Marr's Level 3: L1 & L2 are identical in biology & robotics; differences emerge **only at Level 3**.

1. LEGO Mindstorms EV3 Architecture

- **Processing Unit:** ARM9 @ 300 MHz, 16 MB Flash, 64 MB RAM.
- **Operating System:** Embedded Linux-based OS.
- **Sensor Ports (1-4):** 4 Input ports supporting Analog & high-speed Digital UART up to 460.8 kbit/s.
- **Motor Ports (A-D):** 4 Output ports with built-in optical rotation encoders for position/speed feedback.
- **I/O & Expansion:**
  - High-speed USB (480 Mbit/s) + Host daisy-chaining (up to 3 levels), WiFi dongle, & USB storage.
  - Micro SD-card slot (supports up to 32 GB).
- **Display & UI:** 178×128 monochrome LCD matrix, 6 backlit navigation/debugging buttons.
- **Wireless Comms:** Bluetooth v2.1 + EDR, Apple/Android smart device protocol.

2. Platform Generations & Modules

Platform	CPU / OS	Key Ports & Features
RCX (1998)	Hitachi H8	3 In / 3 Out; Optical IR tower comms
NXT (2006)	ARM7 + AVR	4 In / 3 Out; Bluetooth, 100×64 LCD
EV3 (2013)	ARM9 Linux	4 In / 4 Out; Daisy-chain, USB Host, SD

Sensor & Actuator Modules:

- **Motors:** Large (high torque) & Medium (high RPM) with integrated tacho-encoders.
- **Sensors:** Touch (binary limit), Color (ambient/reflected/RGB), IR Seeker/Beacon, Ultrasonic.

3. Step 1: Avoid Collision (Reflex)

PURE REFLEXIVE BEHAVIOR

- **Objective:** Move continuously; instantly back up and pivot when a physical impact occurs.

START →

Is Collision? (Touch == 1)

YES ↓ | NO ↓

Reverse & Turn
Motors B+C: -30, then steer

Move Forward
Motors B+C: Power = 75

- **Collision == True:** Reverse (-30, 2 rot) → steer turn ($-45/30$, 2 rot).
- Direct hardware-to-motor S-R loop without state memory.

4. Step 2: Collision + Obstacle Avoidance

HIERARCHICAL PRIORITY ARBITRATION

- Combines contact (Touch) and non-contact (Proximity/Force) into a 2-tier reactive hierarchy:

1. Collision? (Touch) | 2. Distance < 40cm?

YES ↓ | YES ↓ | NO ↓

Reverse & Turn
P1: Emergency Reflex

Turn Away
P2: Proximity

Forward
P3: Cruise

- **P1 (Touch):** Emergency physical recovery (overrides all).
- **P2 (Proximity):** Smooth steering turn away before contact.
- **P3 (Default):** Drive straight ahead.

5. Step 3a: 1-Sensor Line Following

BANG-BANG / EDGE TRACKING

- Tracks the **boundary edge** between light and dark surfaces using 1 photo/color sensor.

Reflected Light < Threshold?

YES (On Dark) ↓ | NO (On Light) ↓

Turn Left
Motor C ON / B OFF

Turn Right
Motor B ON / C OFF

- $\text{Ref} < \text{Th}$ (On black) → Steer outside (Turn Left).
- $\text{Ref} \geq \text{Th}$ (On white) → Steer inside (Turn Right).
- **Control Characteristic:** 2-state discrete bang-bang controller; causes high-frequency zigzag oscillation along line edge.

6. Reactive Control Summary

Stage	Sensor Used	Behavior Pattern
Step 1	Touch Switch	Emergency reflex backoff
Step 2	Touch + IR/Dist	Layered priority avoidance
Step 3a	Color/Light	1-Sensor edge tracking

EXAM HIGHLIGHTS: LECTURE 04

EV3 vs NXT/RCX: EV3 introduced Linux OS, ARM9 @ 300MHz, daisy-chaining, USB host, & 4th motor port.
1-Sensor Line Following: Always tracks an **edge** (boundary transition), never the centerline itself.
Priority Arbitration: Safety/impact reflexes always take precedence over non-contact navigation.

1. Control Paradigm Taxonomy

- **Core Philosophy:** How sensing, decision-making, and execution interact.

Paradigm	Information Flow & Focus
Hierarchical / Deliberative	$\text{Sense} \to \text{Plan} \to \text{Act}$ Global world model, top-down plan
Reactive / Behavior-Based	$\text{Sense} \to \text{Act}$ Fast direct S-R loops, no world model
Hybrid	$\text{Plan} \to [\text{Sense} \to \text{Act}]$ Deliberative mission + reactive reflex

Hierarchical (Sense-Plan-Act):

Sense (Observe) → **Plan** (World Model) → **Act** (Execute)

- **'Eyes Closed' Execution:** Robot senses once, computes entire plan, then executes directives without re-evaluating until loop restarts.

2. Monolithic Sensing & Architecture

- **Monolithic Sensing:** All sensor observations fuse into **one global data structure** (World Model) accessed by planner:
  - **A Priori** Pre-loaded maps/building layout.
  - **Sensing Info** Current state & localization ("in NW hallway").
  - **Cognitive** Mission goals (deliver item to Room 118).

Autonomous Vehicle Pipeline:

Stage	Sub-Modules & Tasks
Sensors	Cameras, Radar, LiDAR, GPS / IMU
Perception	Obstacle Detection, Free Space, Localisation
Planning	Route → Behavioral → Trajectory
Control	PID, MPC → Drive-By-Wire (DBW) Actuators

3. Shakey & STRIPS Planning (6-Step Algorithm)

- **Shakey (1967-69, SRI / DARPA):** First AI mobile robot using **Means-Ends Analysis** (reduces state difference Δ).

The 6 Steps in Executing STRIPS

Recursive Stack

1. **Compute difference:** $\Delta = \text{Goal} - \text{Initial State}$. If $\Delta = \varnothing$, terminate.
2. **Select Operator:** If $\Delta \neq \varnothing$, pick 1st operator in Difference Table whose **add-list** negates Δ.
3. **Examine Preconditions:** Check if variable bindings evaluate all preconditions to **TRUE**.
4. **Recurse on Subgoal:** If FALSE, push original goal to stack, set 1st FALSE condition as **new subgoal**, & recurse.
5. **Push to Plan Stack:** When all preconditions match, push operator to plan stack & update world model copy.
6. **Pop & Resume:** Return to parent operator with failed condition to apply it or recurse on next condition.

4. STRIPS Predicates & Operator Anatomy

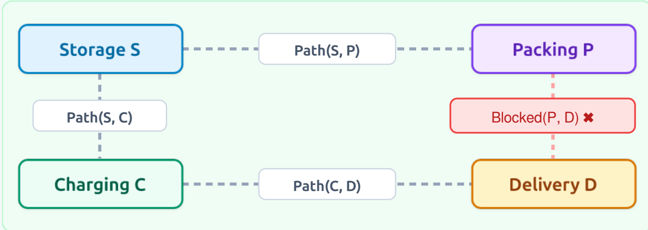
- **Predicates:** UPPERCASE statements evaluating to $\frac{\text{TRUE}}{\text{FALSE}}$ (e.g. $\text{INROOM}(\text{IT}, R_1)$, $\text{CONNECTS}(D_1, R_1, R_2)$).

Operator	Preconditions	Add-List	Delete-List
GOTODOOR (IT, dx)	$\text{INROOM}(\text{IT}, r_k)$ $\text{CONNECT}(d_x, r_k, r_m)$	$\text{NEXTTO}(\text{IT}, d_x)$	—
GOTHRU (IT, dx)	$\text{CONNECT}(d_x, r_k, r_m)$ $\text{NEXTTO}(\text{IT}, d_x)$ $\text{STATUS}(d_x, \text{OPEN})$	$\text{INROOM}(\text{IT}, r_m)$	$\text{INROOM}(\text{IT}, r_k)$

- **Generated Plan:** Stack sequence: $\text{GOTODOOR}(\text{IT}, D_1) \to \text{GOTHRUDOOR}(\text{IT}, D_1)$.

5. Warehouse STRIPS Case Study

Warehouse Layout: Locations Charging (C), Storage (S), Packing (P), Delivery (D).



Operator	Preconditions	Add-List	Delete-List
Move(x, y)	$\text{Path}(x, y)\ \neg\ \text{Blocked}(x, y)$ $\text{At}(\text{Robot}, x)$	$\text{At}(\text{Robot}, y)$	$\text{At}(\text{Robot}, x)$
PickUp(p, x)	$\text{At}(p, x)\ \text{At}(\text{Robot}, x)$ HandEmpty	$\text{Holding}(p)$	$\text{At}(p, x)\ \text{HandEmpty}$
PutDown(p, x)	$\text{Holding}(p)\ \text{At}(\text{Robot}, x)$	$\text{At}(p, x)\ \text{HandEmpty}$	$\text{Holding}(p)$

- **Obstacle Contingency:** Since $\text{Blocked}(P, D)$ is TRUE, planner avoids path $S \to P \to D$ and selects detour:

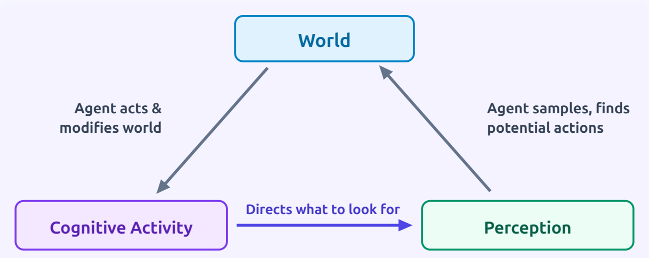
  Detour Plan: $\text{Move}(C, S) \to \text{PickUp}(P_1, S) \to \text{Move}(S, C) \to \text{Move}(C, D) \to \text{PutDown}(P_1, D)$

HIGH-YIELD EXAM DISTINCTION (LECTURE 05)

Deliberative Pros vs Cons: Pro: Computes optimal, goal-directed solutions. Con: Fragile under uncertainty, high computational complexity, slow to react to dynamic changes. **STRIPS Add/Delete Rule:** World state updates **only** by deleting predicates in Delete-List and appending predicates in Add-List upon successful precondition binding.

1. Action-Perception & Ecological Approach

- **Action-Perception Cycle:** Continuous closed feedback loop where perception guides actions, and actions alter the environment.



- **Gibson's Ecological Approach:**
  - Core Tenet *"… the world is its own best representation."*
  - Perception evolved **only to support survival actions** in an ecological niche, not to build abstract 3D maps.

2. Affordances & Perceptual Systems

- **Affordance:** Directly perceivable potentiality for action provided by the environment (e.g. a chair affords **sitability**).
- **Direct Perception:** Requires **no memory, inference, or interpretation** → near-instantaneous execution.
  - Visual: Optic Flow Pattern of apparent visual motion; expanding outward from center gives **time-to-contact** (diving gannets, landing).
  - Non-Visual Sound pitch change when filling a container indicates fullness without knowing cavity dimensions.

Direct Perception (Gibsonian)	Recognition (Cognitive)
Primitive brain structures - Local world models - Fast, lightweight affordances	High cognitive brain - Global world models - Top-down internal templates ("my cup vs your cup")

3. Behavior Acquisition Spectrum

Type	Biological Model	Mechanism & Strategy
Innate	**Arctic Tern** chicks	Born with rule: "Peck at largest red blob" → triggers parent feeding reflex.
Sequence of Innate	**Digger Wasp**	Mating → internal state triggers nest building → visual nest triggers egg laying.
Innate with Memory	**Baby Bees**	Innate flight exploration in expanding arcs to memorize hive appearance and approach vectors.
Learned	**Lion Cubs**	Born with zero hunting behaviors; complex multi-stage skills (stalk, chase) learned over years.

4. Innate Releasing Mechanisms (IRM)

- **IRM Definition:** Theoretical construct where specific sensory triggers (**sign stimuli / releasers**) activate fixed behaviors.
- **Releaser as Control Signal:** Acts as a Boolean latch turning a behavior ON or OFF. If not released, behavior produces zero output.
- **Compound Releasers:** Logical combination of **external stimuli** and **internal motivation** (e.g. $\text{Hungry} \wedge \text{FoodPresent}$).

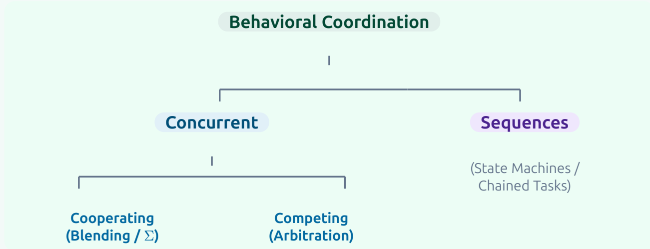
Execution Modes & State Control

Coordination

- **Implicit Chaining:** Execution order emerges dynamically from which releasers are TRUE.
- **Inhibition & FAP Persistence:** Prevents rapid flickering between states (e.g. flee vs feed) by locking behavior for time T.
- **Explicit Sequences:** State-machine order where each behavior runs to completion before advancing.

5. Coordination & Concurrent Behaviors

Behavioral Coordination Architecture:



3 Types of Concurrent Interactions:

- **1. Equilibrium:** Behaviors balance each other out (e.g. **Squirrel** hesitant between food lure and fear of human).
- **2. Dominance (Winner-Take-All):** One behavior completely suppresses others (e.g. sleep vs feed, flight over forage).
- **3. Cancellation:** Opposing stimuli cancel each other, leaving displacement behavior (e.g. **Male Stickleback** fish building a new nest when territory defense & attack cancel).

HIGH-YIELD EXAM DISTINCTION (LECTURE 06)

Design Hierarchy: Always solve behavior using **Affordances** (direct perception) first if possible; resort to **Recognition** (world models) only when cognitive identification is strictly required. **IRM Control vs Perceptual Filter:** An IRM's releaser decides **WHEN** to run; perceptual schema filters **WHAT** to track.

1. Potential Fields Methodology

- **Core Concept:** Most common method for **cooperating behaviors**; outputs are represented as **vectors** in a continuous field.
- **Vector Tuple (m, d):** Written as (m, d) where:
  - m = Magnitude (velocity / force strength).
  - d = Direction (heading angle φ).
- **Field Analogy:** Analogous to magnetic or gravitational fields; the robot behaves as a **charged particle** influenced by surrounding objects.
- **2D (x, y) Grid:** Map divided into square elements; continuous space extends easily from 2D to 3D.

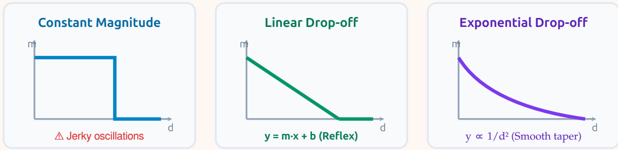
2. Primitive Potential Fields

- **5 Fundamental Primitive Fields:**

Field	Application & Function
Uniform	Constant flow / current across region
Perpendicular	Orient away from walls / corridor centering
Attraction	Target seeking, phototaxis, waypoints
Repulsion	Obstacle avoidance (180° push-away)
Tangential	Orbiting obstacles, perimeter tracing, docking

3. Magnitude Profiles & Decay Functions

- **Magnitude Profile:** Defines how vector magnitude changes over distance x (d).

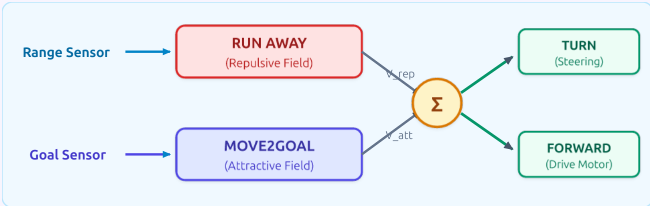


- **1. Constant Magnitude:** Binary velocity inside range; drops abruptly to 0. **Drawback:** causes jerky oscillations on field perimeter.
- **2. Linear Drop-off** $(y = mx + b)$: Response proportional to stimulus (reflexivity). Slows down to prevent overshooting.
- **3. Exponential Drop-off** $(y \propto \frac{1}{d^2})$: Force halves with distance; smooth asymptotic decay eliminates sharp boundaries.

4. Combining Fields & Vector Summation

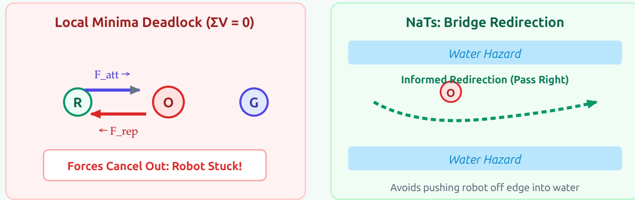
- **Emergent Trajectory:** Emerges from simultaneous linear summation of attractive & repulsive schemas:

Resultant: $V_{\text{resultant}} = \sum_{i=1}^{n} V_i = V_{\text{att}} + V_{\text{rep}}$



5. Local Minima Problem & Solutions

- **Local Minima:** Occurs when opposing forces cancel out $(\sum V = 0)$ before goal is reached, trapping robot in deadlock.



Solution Method	Mechanism & Trade-off
Random Noise	Always-active small noise schema to bump robot out of deadlocks.
Navigation Templates (NaTs)	Uses strategic vector to inform redirection (e.g. avoid water on bridge).
Harmonic Functions	Solves Laplace eq $(\nabla^2 \varphi = 0)$; **guarantees 0 local minima**. Heavy compute.

HIGH-YIELD EXAM DISTINCTION (LECTURE 07)

Pfield Advantages: Intuitive continuous representation, compositional C++ behavioral libraries, extensible to 3D. **Linear vs Exponential Drop-off:** Linear represents reflexivity $(y = mx + b)$ but has boundary edge; Exponential provides smooth decay without jerky chattering. **Harmonic Guarantee:** Only harmonic functions strictly eliminate all local minima traps mathematically.

1. Multi-Robot Systems (MRS) Foundations

- **Core Motivation:** Scaling capability, redundancy, and spatial distribution beyond single-robot limits.

Why Multiple Robots?	Key Challenges & Bottlenecks
· Faster execution & parallelism - Robustness & fault tolerance - Distributed sensing/action - Specialist robot teams	· Communication congestion - Sensor/physical interference - $N \times$ testing & debugging - Emergent unpredictability

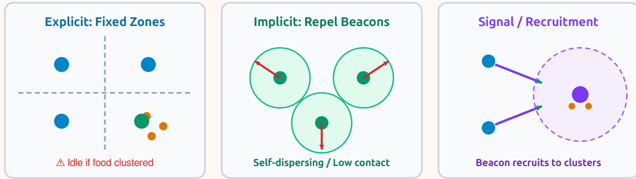
- **Coordination Spectrum:**
  - **Loosely Coordinated** Mapping, exploration, hazardous cleanup (low comms).
  - **Tightly Coordinated** Box pushing, heavy lifting, construction (tight sync).
  - **Task Allocation (MRTA)** Formulated as Multi-Traveling Salesman (mTSP).

2. Core MRS Task Archetypes

Task Archetype	Definition & Scope
Foraging	Collection of randomly placed items.
Consuming	Perform work on object in place (e.g. assemble, disassembly, etc.).
Grazing	Cover entire area adequately (e.g. for lawn mowing, etc.).
Formations / Flocking	Team maintains a geometric pattern while moving.
Object Transport	Collectively moving object.

3. Foraging Dynamics & Distribution

- **Foraging Benchmark:** Speed of completion depends heavily on item distribution (evenly spread vs clustered).



- **1. Explicit (Fixed Zones):** Space divided into rigid areas. Optimal for uniform items; robots become idle if items are clustered.
- **2. Implicit (Repel Beacons):** Omni-directional repulsion keeps robots distributed without central control. Minimizes robot contact.
- **3. Recruitment Signaling:** Robot finding a food cluster turns on beacon to recruits nearby teammates.

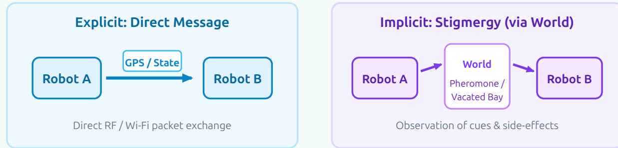
4. Swarm Formations & Flocking (Boids)

- **Reynolds 3 Flocking Rules:** Emergent cohesion vs collision avoidance balance:

- **1. Separation:** Steer to avoid crowding/colliding with local flockmates ($180°$ repulsion).
- **2. Alignment:** Steer towards the **average heading** (v_{avg}) of local flockmates.
- **3. Cohesion:** Steer to move toward the **average centroid** (p_{avg}) of local flockmates.

5. Multi-Robot Communication Taxonomy

Dimension	Taxonomy Categories
Range	None | Near (Local, scalable) | Infinite (Global, heavy)
Topology	Broadcast (all hear) | Addressed | Tree | Graph/Mesh
Bandwidth	High ("free") | Motion-related | Low | Zero



- **Explicit Comms:** Direct message exchange (e.g. drones broadcasting GPS/velocities; "Help, I'm stuck").
- **Implicit Comms (Stigmergy):** Communication through the environment (e.g. sensing markers, seeing a loading bay vacate).

HIGH-YIELD EXAM DISTINCTION (LECTURE 08)

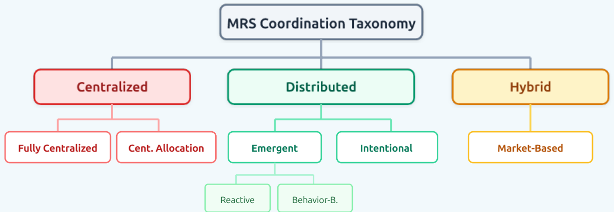
Range vs Societal Performance: Wider communication range is **not** always better; excessive long-range broadcast leads to packet collisions and decreased group performance. **Stigmergy Principle:** "Communicate only what others cannot easily observe themselves." **Repel Scheme:** In clustered foraging, implicit repulsion outperforms fixed partitions by avoiding idle zones.

1. The Coordination Spectrum

- **Coordination Level:** Dictates task decomposition, communication needs, and synchronization tightness.

Coordination Type	Characteristics & Tasks
Loosely-Coordinated	Decomposable into subtasks; independent execution; minimal interaction (e.g. mapping, area search).
Tightly Coordinated	Tasks **not decomposable**; continuous joint execution; high interaction (e.g. box carrying, aerial refueling).

2. MRS Coordination Taxonomy



3. Centralized Approaches (Global Optimum)

- **Core Principle:** Single leader plans for team; attempts global optimal coordination.

Centralized Variant	Mechanism & Trade-offs
Fully Centralized	Leader plans entire team actions as 1 large system. **Pros:** optimal plans. **Cons:** Intractable for $>$ few robots, Single Point of Failure (SPOF), sluggish.
Centralized Allocation	Single agent assigns tasks; robots execute individually. **Pros:** distributed execution. **Cons:** SPOF on allocator.

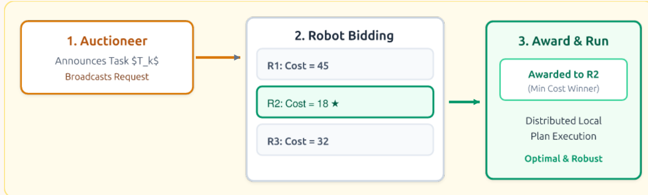
- **Unrealistic Assumptions:** Assumes all info can be sent to leader and world doesn't change during plan construction.

4. Distributed Approaches (Local Autonomy)

- **Principle:** No central authority; each robot plans locally based on sensory perception & neighbor comms.
- **Pros:** Ultra-fast response to dynamics, **no single point of failure**, minimal compute/comms, robust.
- **Cons:** Solutions often **sub-optimal**; cannot handle non-decomposable tasks easily.

5. Sub-Paradigms & Hybrid Market Auctions

Sub-Paradigm	Operation & Example
Emergent (Reactive)	Tight Sense-Act loop ($S \to A$). Fast, simple; cannot plan (e.g. obstacle avoidance).
Emergent (Behavior)	Structured local behaviors with state info. Fast, multi-task, no global plan.
Intentional	Explicit goals & peer negotiation without central planner (e.g. drone trajectory sharing).
Hybrid (Market)	Task auctioneer + distributed bidding based on local cost functions.



HIGH-YIELD EXAM DISTINCTION (LECTURE 09)

Centralized vs Distributed: Centralized yields global optimality but suffers from combinatorial explosion (N-robot intractability) and SPOF; Distributed provides real-time robustness at cost of global optimality. **Intentional vs Emergent:** Intentional robots have explicit goals and deliberate with peers; Emergent swarms exhibit collective behavior solely from reactive local rules. **Market-Based Auctions:** Sweet spot blending efficient centralized task allocation with resilient distributed execution.

1. Market Architecture & Economic Metaphor

- **Market Premise:** Goal points to visit are the **main commodity** exchanged in the market.

- **Team Profit = Sum of Individual Profits:**
  - When individual robots **maximize profit**, the whole team gains.
  - Moves global solution towards optimum without centralized planning.

2. Pricing Dynamics & Specialist Roles

- **Cost:** Based on distance traveled to reach goal.
- **Revenue:** Based on information gained by reaching goal:

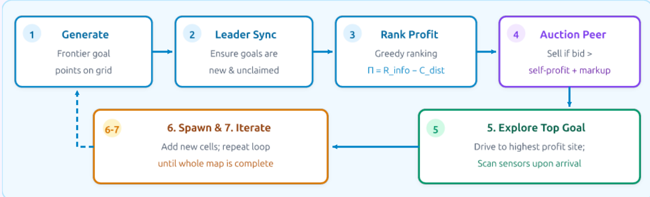
$$R = (\text{\# of unknown cells near goal}) \times (\text{weighting factor})$$

- **Prices Determined by Supply & Demand:** Low-bandwidth mechanisms for communicating aggregate information (map info doesn't need to be communicated repeatedly).
- **Competition vs. Coordination:**
  - **Complementary robots cooperate** (e.g. grasper + transporter offer combined "pick up and place" service).
  - **Similar robots compete**, driving prices down.

3. Grid Exploration & Commodity Model

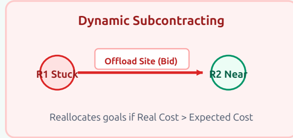
- **Grid Representation:** 2D grid where cells are:
  - **0** Unknown
  - **+1** Occupied
  - **-1** Free Space
- **Main Commodity:** Frontier goal points to explore.
- **Teammate Tracking:** Robots broadcast (x, y, t) coordinates; teammates' footprints are marked as **free space** to prevent obstacle misclassification.

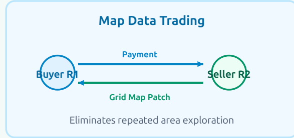
4. Exploration & Auction Algorithm (7 Steps)



Step	Algorithmic Action
1. Generate	Select frontier goal points (greedy, quadtree, random).
2. Sync	Check with leader/peers to ensure goals are unique.
3. Rank	Rank candidate goals greedily by expected profit ($R - C$).
4. Auction	Auction goals; sell if incoming bid > self-profit + markup.
5. Explore	When auctions close, navigate to top-ranked profit goal.
6. Spawn	Sense environment at goal; spawn new frontier cells.
7. Iterate	Repeat cycle until entire map is completely explored.

5. Replanning, Subcontracting & Trading





- **Dynamic Subcontracting:** When unforeseen obstacles cause $C_{\text{real}} \gg C_{\text{expected}}$ or a robot fails, expensive sites are re-auctioned.
- **Map Data Trading:** Robots sell discovered map patches to peers, preventing duplicate scanning.

6. Experimental Validation & Results

- **Testbed:** 4–5 robots (FOG gyros, 16 sonars) across cluttered rooms, patio, and dynamic conference hall (100 people).
- **Performance Metric:** Exploration efficiency:

$$\text{Exploration Efficiency} = \frac{\text{Area Covered } [m^2]}{\text{Distance Traveled } [m]}$$

- **Empirical Result:** Market coordination improved efficiency over uncoordinated exploration by a factor of **3.4**×.

HIGH-YIELD EXAM DISTINCTION (LECTURE 10)

Why Market Approaches Excel: Blends centralized optimality (Pareto team gain) with distributed fault tolerance (task re-bidding on robot failure). **Price as Information Filter:** Prices convey compact aggregate utility without broadcasting heavy raw maps. **Efficiency Factor:** $3.4\times$ empirical gain in area covered per meter traveled.